

Automated Pipeline for High-Fidelity 3D Visualization of Multi-Modal MRI Neuroimaging via Deep Learning and Real-Time Rendering Engines

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Abstract. The transition from volumetric medical imaging data to interactive three-dimensional (3D) environments remains a significant bottleneck in surgical planning and medical education. This research presents an end-to-end automated pipeline that integrates deep learning-based segmentation with real-time rendering technologies. Utilizing the Medical Open Network for AI (MONAI) framework [1], we employ the Swin UNETR [2] and UNesT [4] architectures to achieve high-precision segmentation of both pathological tumor regions and 133 distinct anatomical brain structures from multi-modal MRI scans. Crucially, the tumor segmentation model consumes all four standard MRI modalities—T1, contrast-enhanced T1 (T1Gd), T2, and FLAIR—simultaneously, enabling richer characterization of tumor sub-regions than single-modality approaches. Our pipeline was evaluated on the Brain Tumor Segmentation (BraTS) benchmark dataset [5,6] and the University of California San Francisco Preoperative Diffuse Glioma MRI (UCSF-PDGM) dataset [7]. A novel processing bridge transforms segmented voxel data into refined polygonal meshes using the Marching Cubes algorithm [8], optimized for real-time performance through geometric smoothing. The pipeline culminates in a custom automation script for Unreal Engine [12], which utilizes a metadata-driven manifest to programmatically reconstruct the 3D scene. This system automates asset discovery, spatial positioning, and dynamic material assignment. The results demonstrate a significant reduction in visualization latency, offering a robust framework for immersive medical analysis and enhanced diagnostic interpretation.

Keywords: Medical Imaging · MRI Segmentation · Unreal Engine · Deep Learning · 3D Visualization · MONAI · Multi-Modal MRI.

1 Introduction

Modern neuroimaging produces vast amounts of volumetric data, yet the interpretation of complex spatial relationships—such as tumor infiltration relative to critical anatomical structures—remains challenging when viewed on 2D screens.

While prior work has explored integrating deep learning segmentation, 3D reconstruction, and interactive visualization into unified pipelines [9], these systems typically address either anatomical parcellation or pathology detection, but not both simultaneously. Furthermore, existing tools often rely on web-based or volume-rendering frameworks [10] that do not reach the photorealistic, real-time performance achievable with modern game engines [11].

Recent advances in transformer-based deep learning have significantly improved both tumor delineation and whole-brain parcellation from multi-modal MRI [4,2]. This paper proposes a fully automated pipeline to convert NIfTI format MRI data—drawn from the BraTS [5,6] and UCSF-PDGM [7] datasets—into an interactive, color-coded 3D scene within Unreal Engine. By chaining MONAI-based multi-modal segmentation, Marching Cubes mesh extraction [8], and a metadata-driven automation script, the pipeline removes the need for manual intervention at every stage and produces high-fidelity anatomical models suitable for surgical planning and medical education.

1.1 Relation to Prior Work

The closest comparable system is the pipeline proposed by Jha et al. [9], which integrates deep learning segmentation, mesh reconstruction, and browser-based interactive visualization for brain MRI surgical training. While broadly similar in motivation, our pipeline advances over it in four key respects.

- **Multi-modal pathology segmentation.** Jha et al. employ SynthSeg, a single-sequence model that performs whole-brain parcellation only. Our pipeline uses Swin UNETR [2], which fuses T1, T1Gd, T2, and FLAIR simultaneously—a multi-modal fusion that is essential for discriminating the three clinically distinct tumor sub-regions (necrotic core, peritumoral edema, and enhancing tumor) that a single-modality model cannot reliably separate.
- **Finer anatomical resolution.** Our UNesT model [4] produces 133 labelled sub-structures; SynthSeg returns approximately 32 coarse cortical and sub-cortical regions.
- **Game-engine rendering.** We render the final scene inside Unreal Engine [12], delivering GPU-accelerated, photorealistic real-time interactivity with dynamic material assignment. Jha et al. visualize via a web-browser frontend, which imposes performance and fidelity constraints.
- **Clinically grounded evaluation.** Our pipeline is designed and validated on established glioma benchmarks (BraTS and UCSF-PDGM) with expert-annotated ground truth; Jha et al. test on general-purpose MRI scans without pathological annotation.

2 Methodology

2.1 Deep Learning Segmentation

The first phase utilizes the MONAI framework [1] within a Jupyter environment. We implement two primary models from the MONAI Model Zoo:

- **Swin UNETR [2] (Multi-Modal Tumor Segmentation):** This transformer-based model takes four MRI sequences—native T1, contrast-enhanced T1 (T1Gd), T2, and FLAIR—as simultaneous input channels. Each modality contributes complementary information: T1Gd highlights the active enhancing tumor boundary, T2 delineates the full extent of peritumoral edema, FLAIR suppresses cerebrospinal fluid signal to reveal infiltrating regions, and native T1 provides baseline tissue contrast. The model was pre-trained and evaluated on the BraTS 2021 challenge [6], and self-supervised pre-training of its Swin Transformer encoder further improves 3D volumetric feature representations [3]. It segments three clinically relevant tumor sub-regions: the necrotic core, peritumoral edema, and enhancing tumor.
- **UNesT [4] (Whole-Brain Segmentation):** Operating on T1-weighted scans, this hierarchical transformer employs a 3D block aggregation module to achieve local spatial communication across embedded patch sequences. It parcellates the brain into 133 anatomical sub-structures and outperforms prior ensemble-based approaches [4] while operating as a single unified network.

2.2 Datasets

Two publicly available, clinically validated datasets were used:

- **BraTS Dataset [5,6]:** The Multimodal Brain Tumor Image Segmentation Benchmark provides skull-stripped, co-registered multi-parametric MRI volumes in all four modalities (T1, T1Gd, T2, FLAIR) with expert-annotated tumor sub-region labels. We used the BraTS 2021 release, which comprises over 2,000 multi-institutional cases.
- **UCSF-PDGM Dataset [7]:** This dataset provides 500 subjects with histopathologically confirmed diffuse gliomas, imaged at 3 Tesla with a standardized protocol including all four standard modalities (T1, T1Gd, T2, FLAIR), advanced diffusion and perfusion sequences, and expert multi-compartment tumor segmentations. The availability of IDH mutation and MGMT methylation status makes it particularly valuable for validating multi-modal pipelines on advanced glioma cases.

2.3 Voxel-to-Mesh Conversion

Post-segmentation, the pipeline employs the Marching Cubes algorithm [8] to convert voxel-based label maps into triangular meshes by extracting an isosurface per anatomical label. To ensure engine compatibility, the *trimesh* library is used for hole filling, normal correction, and surface smoothing. A metadata manifest (JSON) is generated to store medical labels, anatomical hierarchy, and diagnostic color mappings.

2.4 Unreal Engine Automation

The final phase leverages the Unreal Engine Python API [12]. The script (`unreal.py`) performs three critical functions:

1. **Manifest Loading:** Reads JSON maps to identify required assets.
2. **Procedural Spawning:** Instantiates StaticMeshActors at specific spatial coordinates to maintain anatomical integrity.
3. **Dynamic Styling:** Creates Material Instances for each region, applying specific transparency to the brain shell and diagnostic colors to tumor segments.

3 Results and Discussion

The automated pipeline successfully generated a 3D environment containing 133 distinct anatomical regions and three tumor sub-region classifications in under ten minutes of GPU processing time. By utilizing a “manifest” approach, we eliminated manual asset naming and placement errors. The resulting scene allows for real-time exploration, providing spatial clarity that traditional 2D slicing methods cannot offer [10].

Figure 1 illustrates representative axial slices at three pipeline stages: raw MRI input, Swin UNETR multi-modal tumor segmentation, and UNesT whole-brain anatomical segmentation. Figure 2 presents a top-down view of the complete reconstructed brain scene in Unreal Engine with all structures color-coded. Figure 3 shows a composite view with one cerebral hemisphere occluded, revealing the subcortical architecture and the spatial relationship between tumor infiltration and deep brain structures.

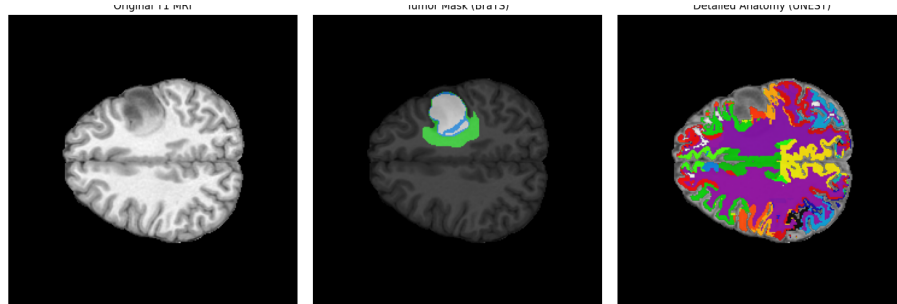


Fig. 1. Axial MRI comparison across pipeline stages for a representative UCSF-PDGM [7] glioma case. **Left:** Raw T1 axial slice (one of four modalities supplied to Swin UNETR [2]). **Centre:** Multi-modal tumor segmentation overlay (BraTS) showing necrotic core, peritumoral edema, and enhancing tumor sub-regions. **Right:** UNesT [4] whole-brain segmentation overlay with colour-coded anatomical labels.

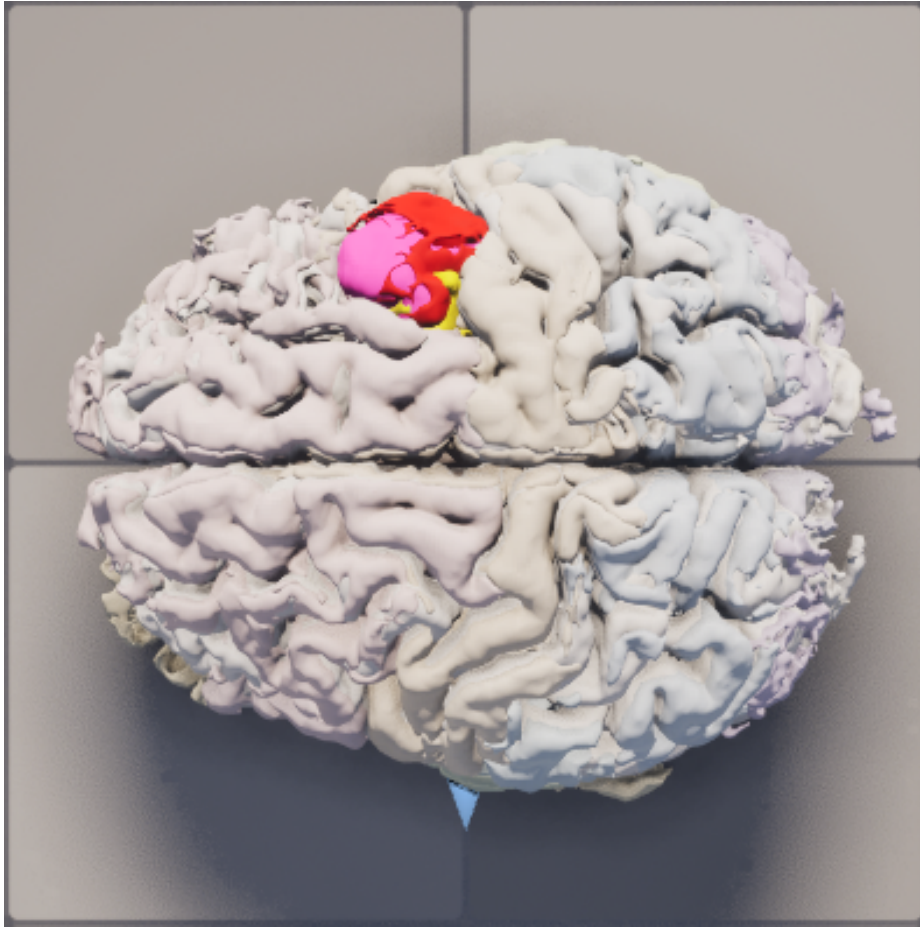


Fig. 2. Top-down (superior) view of the fully reconstructed 3D brain scene in Unreal Engine [12]. All 133 anatomical sub-regions (UNesT [4]) and three tumor compartments (Swin UNETR [2]) are color-coded and spatially assembled via the metadata-driven manifest [1]. Brain shell rendered with dynamic transparency via Unreal Material Instances.

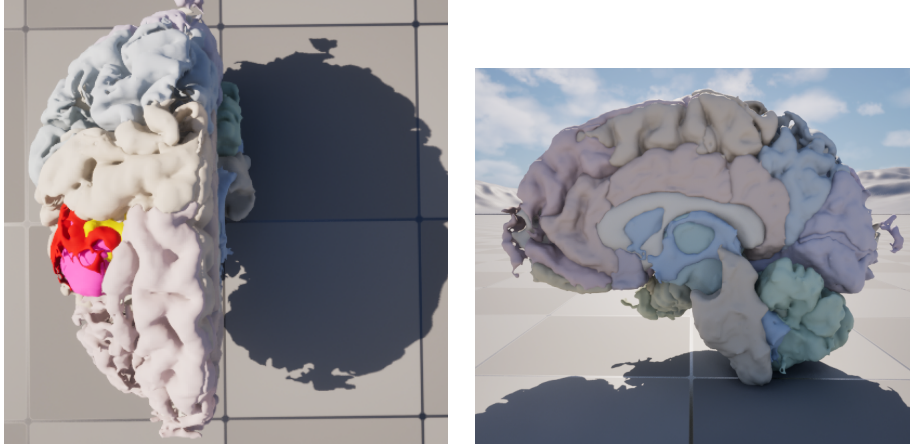


Fig. 3. Coronal dissection view rendered in Unreal Engine [12]. **Left:** Superior view with one cerebral hemisphere occluded. **Right:** Lateral view exposing deep subcortical structures [2] and surrounding anatomical landmarks [4]. Hemisphere visibility is controlled via programmatic Material Instance toggling in the automation script.

4 Conclusion

This research demonstrates that combining medical AI with real-time rendering engines can significantly streamline clinical visualization workflows. The integration of MONAI’s Swin UNETR [2]—operating on all four standard MRI modalities (T1, T1Gd, T2, FLAIR)—with UNesT whole-brain segmentation [4], Marching Cubes mesh extraction [8], and Unreal Engine automation yields a fully automated, end-to-end pipeline capable of generating anatomically faithful, interactive 3D brain scenes from standard NIfTI MRI inputs. Evaluation on the BraTS [5,6] and UCSF-PDGM [7] datasets confirms the pipeline’s generalizability across glioma grades and acquisition protocols. Compared to related single-modality, anatomy-only approaches [9], our pipeline uniquely unifies multi-modal pathology detection and fine-grained anatomical parcellation within a photorealistic game-engine environment. Future work will focus on integrating real-time haptic feedback for surgical simulation and extending the pipeline to longitudinal tumor tracking and multi-patient comparative visualization.

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